Akoizo Original Practice - Form 4 - Answer key

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Wheels are 6.0 cm in diameter. How far does one axle turn move the vehicle?

A. 12.0 cm

B. 37.7 cm

C. About 18.8 cm

D. 6.0 cm

Correct: C

2. (2 points) If vehicle speed doubles, how does its kinetic energy change?

Unit: Energy from a falling mass.
Worked-case data: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?
Separate investigation observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

A. It halves

B. It stays the same

C. It becomes four times larger

D. It doubles

Correct: C

3. (2 points) What must be detached when impounding?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. The falling mass

B. The wheels

C. The stopping mechanism

D. The egg holder

Correct: A

4. (2 points) What happens if the egg breaks on the first run?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

A. A violation, and no second run

B. Only a time penalty

C. Nothing

D. A 300-point penalty

Correct: A

5. (4 points) Why does a bigger distance ratio reduce acceleration?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

answer: The same force on the string becomes a smaller force at the wheel, since force divides by the ratio. (4 points).

6. (4 points) What violations come from the egg and barrier?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

answer: Breaking the egg, any part besides the egg touching the barrier, and the egg tip going past the barrier's front. (4 points).

7. (4 points) Calculate the arc length for a chord of 8.00 m and a radius of 11.0 m.

answer: θ = 2 asin(4.00 ÷ 11.0) about 0.744 rad; arc = 11.0 x 0.744 about 8.19 m. (4 points).

8. (4 points) How should a team find the largest error source?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

answer: Run many tests at one setting, measure the spread, and test each suspected cause separately. (4 points).

9. (4 points) How much kinetic energy does a 0.80 kg vehicle have at 2.5 m/s?

answer: ½ x 0.80 x 2.5^2 = 2.5 J. (4 points).

10. (4 points) Why test on a floor similar to the tournament's?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

answer: Floor surfaces change traction, rolling resistance, and skid, shifting stopping distances. (4 points).

11. (6 points) Solve this case and explain your reasoning: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.

reasoning: a = v^2 ÷ (2d) = 1.5^2 ÷ (2 x 0.06) = 2.25 ÷ 0.12 about 18.8 m/s^2.
In g: 18.8 ÷ 9.8 about 1.9 g.
This is gentle; the egg should survive. (3 points).

conclusion: About 1.9 g, a safe stop. (3 points).

12. (2 points) Why did Run 2 travel less?

Supplied synthetic observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

A. The mass was lighter

B. Pile-up increased the effective axle radius

C. The wheels shrank

Correct: B

13. (2 points) Define “Target Point” in the context of Scrambler.

answer: Point 7.00-10.00 m from the Start Point. (2 points).

14. (2 points) Define “Bonus Line” in the context of Scrambler.

answer: Line halfway to the target where the cans sit. (2 points).

15. (2 points) Define “Threaded rod” in the context of Scrambler.

answer: Screw that moves a nut a fixed distance per turn. (2 points).

16. (2 points) Define “Launcher” in the context of Scrambler.

answer: Separate stand that holds and drops the mass. (2 points).

17. (2 points) Define “Impound” in the context of Scrambler.

answer: Turning in the vehicle before the event. (2 points).

18. (2 points) Define “g-force” in the context of Scrambler.

answer: Acceleration in multiples of 9.8 m/s^2. (2 points).